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1.0 - Executive Summary

Process Flowsheet Simulation using Excel has the objective to demonstrate and calculate the production of Cyclohexane using Excel and VBA coding. The production process occurs via the hydrogenation of Benzene in a bubble column reactor with the assistance of a nickel catalyst.

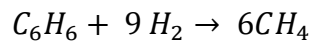
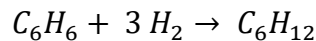
It is possible to determine the conversion and selectivity of the Benzene during the cycle, as well as the purge percentage and all the compositions from the streams, including the purity of the product Cyclohexane.

After performing an optimization, the results were found to be 99.65% conversion with 99.7% selectivity. The Cyclohexane produced was determined to be 99.64% pure, and this process was carried at 32.77 KMTA with a FCI of \$43.92MM (USD). To scale up to a 200 KMTA plant, the FCI was calculated to be \$152.13MM (USD). More details and discussion about those results can be found in the Conclusion and Analysis part of this report.

2.0 – System and Procedure

The system includes a bubble column reactor, followed by two separation columns. More details can be found in the Process Flow Diagram in Appendix II.

Benzene enters as a fresh feed at 100°F and 15 Psia by a supply stream (100 lbmol/h), followed by a secondary stream of fresh feed (313 lbmol/h) containing 97.5 % hydrogen gas (305.175 lbmol/h), 0.5% nitrogen gas (1.565 lbmol/h) and 2% methane gas (6.26 lbmol/h) at 120°F and 335Psia. The two streams are combined with two recycling streams from the first high pressure separator, and all streams are mixed inside a mixer. The mix is then fed to the reactor where two reactions take place at 400°F and 315 Psia:



The resultant stream from the reactor is then cooled to 120° F and fed to the high-pressure separator at 315 Psia, where a vapor stream comes out of the top and a liquid stream comes out of the bottom. The top stream is followed by a purge and a recycle stream back to the mixer. The bottom stream is also split, where the resulting streams will be a liquid recycle back to the mixer and a secondary stream as the feed for the low-pressure separator. At the low-pressure separator, the final product will be collected out of the bottom stream, which may also include some impurities. The top vapor stream is composed mainly of methane.

The system presents some parameter that must be met in order to be considered a successful result. The vapor recycled stream must be adjusted in order to maintain a hydrogen to Benzene molar ration of 3.3 in the mixer, while the liquid recycled stream must be no more than 45% of the fresh feed liquid Benzene flow rate.

The procedure was completed by the following steps:

- Develop a process flow diagram.
- Material Balance, including the determination of the rate of reactions, $3i$
- Determination of the tear streams.
- Determination of the vapor/liquid split (of both the separators).
- Determination of the purge and the recycle fractions.
- Setting up equations and VBA code.
- Optimizing the system to meet the requirements.

In the first iteration, it was assumed only pure fresh feed and no recycle feed was input into the reactor. Starting on the second iteration, some recycle was added to the fresh feed. The iterations were repeated until a steady state was reached.

3.0 – Physical Properties

Table 1: Physical properties of components.

Property Name	Units	H2	N2	CH4	C6H6	C6H12_1
Molecular Weight		2.0160	28.0130	16.0430	78.1140	84.1620
Critical Temperature	C	-239.9500	-146.9500	-82.5500	288.9500	280.2500
Critical Pressure	atm	12.8000	33.5000	45.4000	48.3000	40.2000
Critical Volume	cm ³ /mol	64.9115	89.6408	99.2094	258.7791	308.3679
Compressibility Factor		0.3050	0.2900	0.2880	0.2710	0.2730
Accentric Factor		-0.2200	0.0400	0.0080	0.2120	0.2130
Normal Boiling Point	C	-252.7500	-195.7500	-161.4500	80.1500	80.7500
Freezing Point	C	-259.1500	-209.8500	-182.4500	5.5500	6.5500
Liquid Molar Volume	cm ³ /mol	28.6127	34.7065	37.7482	95.5084	116.6608
Enthalpy of Formation	kcal/mol	0.0000	0.0000	-17.8900	19.8200	-29.4300
Gibbs Energy of Formation	kcal/mol	0.0000	0.0000	-12.1500	30.9900	7.5900
Heat of Vaporization	kcal/mol	0.2160	1.3330	1.9550	7.3520	7.1600
Heat Capacity	cal/(mol C)					
A1		6.1038	7.4400	4.5980	-8.1010	-13.0270
A2 x 1.0E3		3.5399	-3.2400	12.4500	113.3000	146.0000
A3 x 1.0E6		-5.3789	6.4000	2.8600	-72.0600	-60.2700
A4 x 1.0E9		4.2220	-2.7900	-2.7030	17.0300	3.1560
A5 x 1.0E12		-1.4176	0.0000	0.0000	0.0000	0.0000
A6 x 1.0E15		0.1704	0.0000	0.0000	0.0000	0.0000
Low Temperature Limit	C	-73.1500	26.8500	26.8500	26.8500	26.8500
Upper Temperature Limit	C	2726.8500	936.0500	1367.8500	1137.3500	1082.3500
Liquid Vapor Pressure	Pa					
A1		25.3527	46.1539	43.3183	73.8624	73.3379
A2		-142.8939	-984.2006	-1380.5780	-5970.4390	-5885.9268
A3		0.0000	0.0000	0.0000	0.0000	0.0000
A4		0.0971	0.0232	0.0116	0.0055	0.0055
A5		-2.9194	-5.4530	-4.3964	-8.0798	-8.0304
A6		0.0000	0.0000	0.0000	0.0000	0.0000
A7		6.0000	6.0000	6.0000	6.0000	6.0000
Low Temperature Limit	C	-258.6136	-209.8500	-182.4500	20.2109	19.5774
Upper Temperature Limit	C	-239.9500	-146.9500	-82.5500	288.9500	280.2500
Transport Properties						
Diffusion Coefficient	m ² /s					
Mass Transfer Coefficient	m/s					
Heat Transfer Coefficient	kcal/(m ² s C)					
Viscosity	cp					
Thermal Conductivity	kcal/(m s C)					
Solid State		0.0	0.0	0.0	0.0	0.0

Extents of reaction (ζ_1 and ζ_2) were calculated based on the conversion rate and selectivity of the Benzene per pass, which was used to calculate the material balance for each component. The extent of reaction only became constant after the system converged, which means the system reached steady state.

For the separation columns, the Rachford Rice equation² was used to obtain the vapor and liquid feed values. This was a reasonable assumption since the separation column works in this case as a flash distillation, where the lighter stream (vapors) comes up from the top, and the heavier stream (liquids) comes off from the bottom. It is also known that at the split point. The composition of the streams will be the same, only varying the flow rate involved after the split.

4.0 – Components

For this process the followed components were involved:

- H_2 – hydrogen gas
- N_2 – nitrogen gas
- CH_4 – methane
- C_6H_6 – Benzene
- C_6H_{12} – Cyclohexane

The composition for each component in relation with its stream can be found in the table below.

Table 2: Component stream flows.

Hydrogen	S0H	S0B	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
[lbmol/h]	305.175	0	330.51	29.75	29.25	0.5	4.07	25.18	0.15	0.35	0.35	0.003

Nitrogen	S0H	S0B	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
[lbmol/h]	1.56	0	9.51	9.51	9.08	0.43	1.26	7.81	0.13	0.30	0.30	0.007

Methane	S0H	S0B	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
[lbmol/h]	6.26	0	38.16	39.96	35.48	4.48	4.93	30.54	1.36	3.12	2.93	0.19

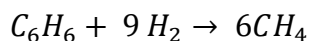
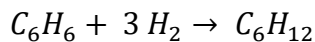
Benzene	S0H	S0B	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
[lbmol/h]	0	100	100.15	0.5	0.005	0.5	0.0007	0.005	0.15	0.35	0.005	0.34

Cyclohexane	S0H	S0B	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
[lbmol/h]	0	0	44.5	143.9	1.52	142.3	0.21	1.31	43.21	99.14	1.49	97.65

5.0 – Equipment

The following equipment were used in the process of Cyclohexane production:

- Reactor: vessel where the chemical reactions took place.
Chemical reactions occurred at 400°F and 315 psia



- High pressure separator: vessel where the first separation took place.
Separation occurred at 120°F and 310 psia, with a overhead gas purge fraction of 0.1391 and a bottom liquid recycle fraction of 0.3035.
- Low pressure separator: where the final separation took place, and the final product was collected.
Separation occurred at 15 psia with a product flow rate of 32.77 KMTA.

6.0 – Discussion and Analysis

After 103 iterations the system converged, which means the flow rate and composition of the stream S2 matched the tear stream based on the concept $y = g(y)$.

The logic used for solving this problem was to consider the first iteration as a straight iteration without any recycling streams coming into the mixer, followed by the second iteration where the recycling streams were present. By using the Direct Substitution Method, a VBA code was written in order to find the point where the S2 stream and the tear stream shared the same value.

After the system converged, Solver was used to optimize the system by implementing the specifications needed: a molar ratio fraction of 3.3 for hydrogen – Benzene into the mixer and a liquid recycle stream that should be no more than 45% of the fresh Benzene flow rate.

It was determined that the system operates at 99.65% overall Benzene conversion and 99.70% overall Benzene selectivity. The product collected at 32.77 KMTA presents a 99.64% purity, which is above the 99% requirement by the industry demand.

In order to calculate the production of 200KMTA flow rate of product, the scale factor was modified from 1.0 to 6.1025. This production gives a Fixed Capital Investment (FCI) of \$142.31 MM(USD) in 2016, to convert this value to 2019 it was taking into consideration an average inflation of 2.25%/ year, which gives a total of \$152,123,767.14 US Dollars or \$152.13MM(USD).

In order to change the process to remove the liquid recycle for the 200KMTA Cyclohexane production, the high-pressure bottom liquid recycle fraction was set to zero, which means all the bottom S5 stream will be the feed for the low-pressure separator S9 stream. After the optimization of this process, the scale factor was changed to 6.1014, and the high-pressure overhead gas purge fraction was set to 0.1540. By doing those changes, the purity of the product went from 99.64% to 99.50%, the overall Benzene conversion went from 99.65% to 99.51%, while the overall Benzene selectivity and the FCI (Fixed Capital Investment) did not change.

7.0 – Appendix

7.1 – Appendix I

The following are formulas used during this project:

- Mass Balance
 Hydrogen: $n_{H_2} = n_{H_2}^0 - 3 * 3_1 - 9 * 3_2$
 Nitrogen: $n_{N_2} = n_{N_2}^0$
 Methane: $n_{CH_4} = n_{CH_4}^0 + 6 * 3_2$
 Benzene: $n_{C_6H_6} = n_{C_6H_6}^0 - 1 * 3_1 - 1 * 3_2$
 Cyclohexane: $n_{C_6H_{12}} = n_{C_6H_{12}}^0 + 1 * 3_1$
- Conversion of Benzene
 $3_1 + 3_2 = n_{C_6H_6}^0 * \text{conversion}$
- Rachford-Rice Equation²

$$x_i = \frac{z_i}{1 + \frac{V}{F}(K_i - 1)} \quad \text{and} \quad y_i = K_i * x_i$$

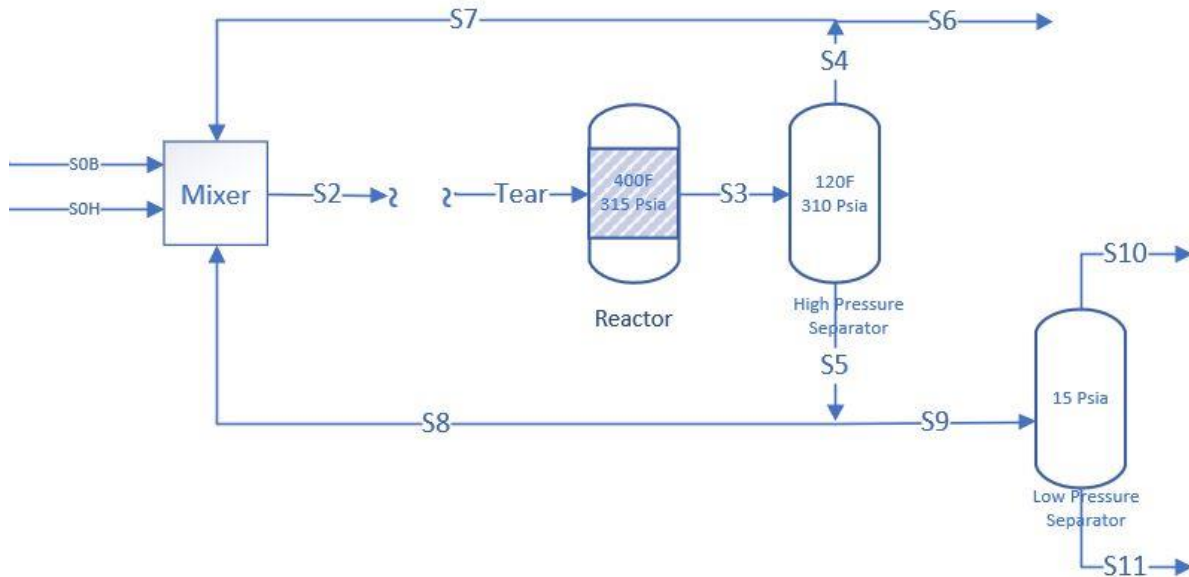
Where x_i is the fraction of the i component of the liquid stream, z_i is the fraction of the feed for the i component and y_i is the fraction of component i of the vapor stream. K_i is the value K 's for the i component, for both the high and low pressure.

- Fixed Capital Investment (FCI)¹

$$FCI = 0.017 * (\text{Number of Functional Units}) * 1000 * (\text{Product Flow Rate})^{0.65}$$

7.2 – Appendix II

Figure 1: Process Flow Diagram used to model the production of Cyclohexane.



S0B: Fresh Benzene Feed Stream

S0H: Fresh Hydrogen Feed Stream

S2: Combined Feed for the Reactor

S3: Combine Product from the Reactor

S4: Overhead Vapor Stream from High-Pressure Separator

S5: Bottom Liquid Stream from High-Pressure Separator

S6: Purge Stream

S7: Vapor Recycle Stream from High-Pressure Separator

S8: Liquid Recycle Stream from High-Pressure Separator

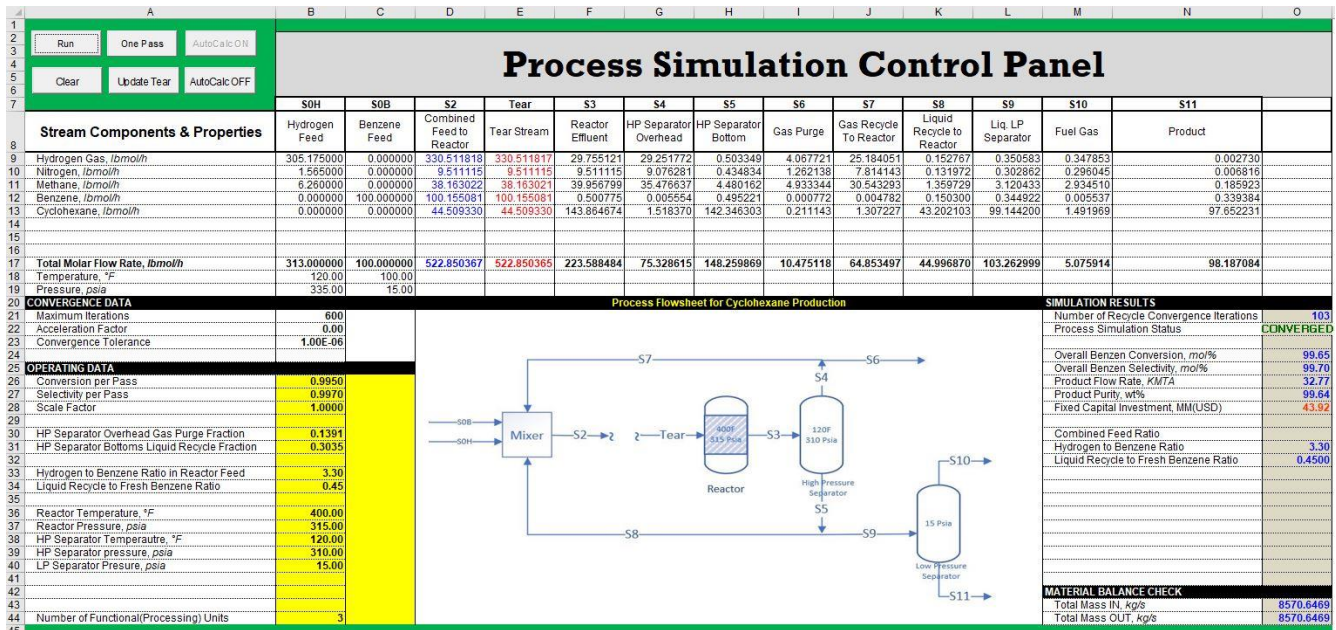
S9: Combined Feed for the Low-Pressure Separator

S10: Overhead Vapor from Low-Pressure Separator

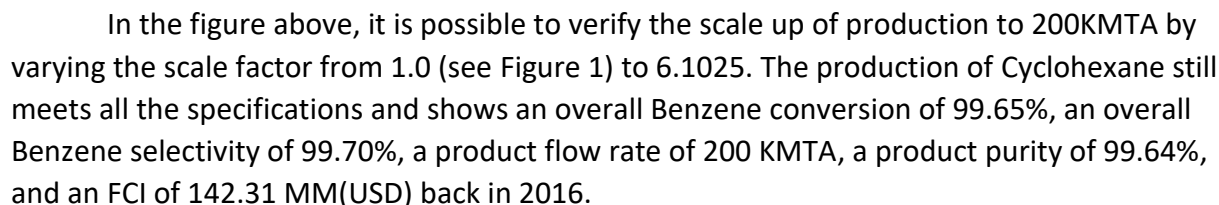
S11: Bottom Liquid Stream from Low Pressure Separator (Final Product)

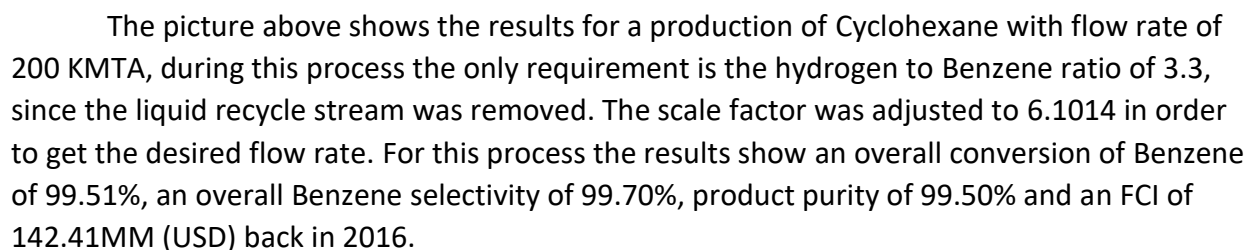
7.3 – Appendix III

Figure 2: Results of Part 1 – 4



In the figure above, it is possible to verify the production of Cyclohexane meets all the specifications. After optimization using Solver, the following results were obtained: a purity of 99.64%, a production of 32.77 KMTA, an overall conversion of Benzene of 99.65%, an overall selectivity of the Benzene of 99.70%, and an FCI of 43.92MM (USD) in 2016.





7.0 – References

- 1) Chi Zhang, & Mahmoud M El-Halwagi. (2017). *Estimate the capital cost of shale-gas monetization projects*. Chemical Engineering Progress, 113(12), 28-32. Retrieved from <https://search.proquest.com/docview/1979759841>
- 2) Rachford Jr, H. H., & Rice, J. D. (1952). *Procedure for use of electronic digital computers in calculating flash vaporization hydrocarbon equilibrium*. Petroleum Transactions, AIME, 195, 327-328.